

Data Vizards

Introduction

This project explores patterns in global mutual-fund performance using a dataset sourced from Morningstar, one of the world's leading providers of investment research and analytics.

Our objective was to examine how geographic domicile, investment style, and market capitalization influence long-term fund returns and risk. The dataset contains over 500 equity funds with the following key variables:

Variable	Type	Description
Domicile	Categorical	Country or region where each fund is based
Investment Style (Box)	Categorical	Market-cap and strategy classification
Industry Category	Categorical	Sector or investment theme
Total Return (6 months - 10 Years)	Numeric	Percent return over a five-year period
Risk Rating	Numeric	Morningstar-assigned overall risk score

The story we sought to tell is that fund size and domicile meaningfully affect performance dispersion and risk exposure. Large-cap funds tend to offer stable but moderate returns, while smaller-cap funds show higher variance, indicating greater opportunity and risk. Geography, meanwhile, acts as a hidden variable shaping the style composition of portfolios.

Exploratory Analysis

Our exploratory work began by cleaning the dataset. Removed incomplete records and normalized categorical labels. We produced simple histograms and scatterplots to get a feel for the data in early visual checks. The histogram of 5-Year Returns showed a right-skewed distribution, suggesting a few very high performers. A scatterplot of Market Cap vs Return revealed a loose positive trend, confirming the expected trade-off between risk and reward. Cross-tabulations of Domicile × Style highlighted the dominance of U.S. funds in nearly every style class.

From these observations, we narrowed our focus to four perspectives:

1. Spread of returns by investment style (volatility dimension).
2. Sector based concentration by domicile.
3. Relative performance of 6months to a year across Sectors.
4. Joint relationship of risk, return, and industry category.

Visualization 1 – Violin Plot (Distribution of Returns by Investment Style)

Variables Mapped:

- x-axis: Investment Style box
- y-axis: 5-Year Return (%)

Refinement Process:

The original boxplot version lacked nuance in showing intra-category variation.

We converted it to a violin plot to reveal distribution shape and outliers more clearly.

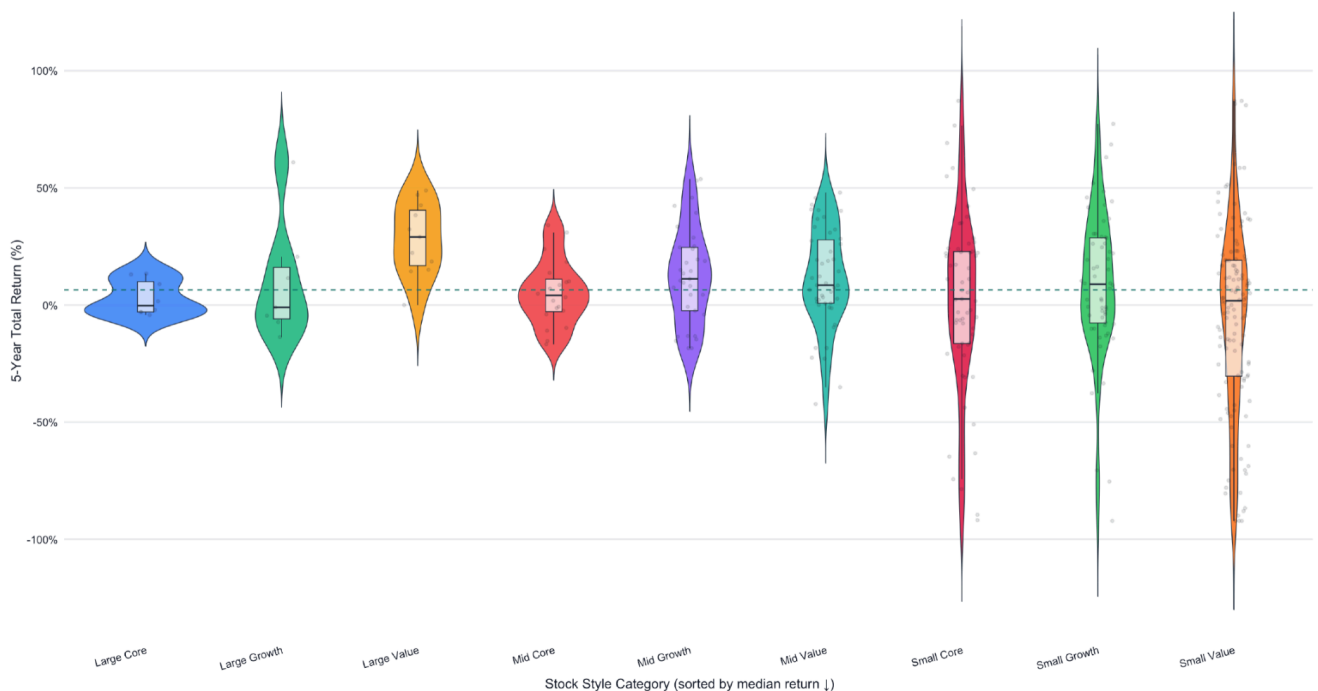
Ordered styles from Large Value → Small Growth to emphasize the increasing spread.

Insights:

Small-cap Growth and Small-cap Value funds display the widest spread of returns, highlighting greater volatility. Large Core and Large Value funds cluster tightly around the median, indicating predictable, stable performance.

Distribution of 5-Year Returns by Category

Chart information - Violin: full distribution | dashed: overall median



Visualization 2 – Mosaic Plot (Domicile vs Investment Style)

Type: Proportional area plot

Variables Mapped:

- x-axis: Domicile
- y-axis: Investment Style (Box)
- tile area: Number of funds

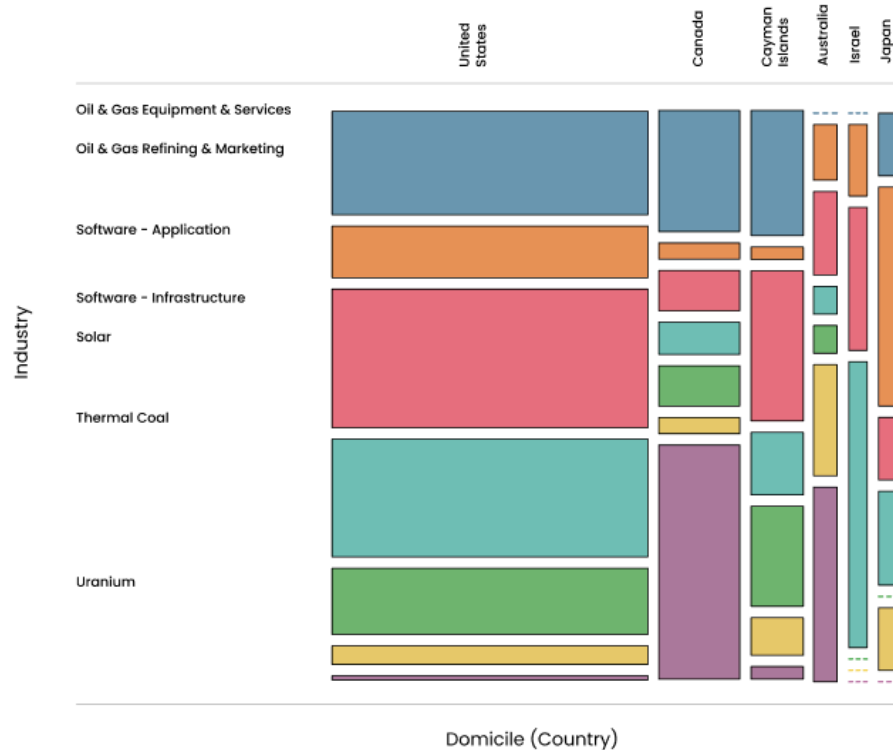
Refinement Process:

The first draft included all 20 + domiciles, cluttering the display.

We filtered to the top six domiciles, replaced the grayscale palette with distinct colors, and the x-axis was reordered by box size.

Insights:

The U.S. dominates across nearly every style, especially Large Core and Large Grow



Visualization 3 – Heat Map (Return by Industry)

Type: Matrix plot with color intensity

Variables Mapped:

- x-axis: Industries
- y-axis: Time Period (1Y, 2Y, 5Y, 10Y)
- Area: Total Return

Refinement Process:

We initially visualized this data as a line graph but switched to a matrix format to make trends more visually appealing and understandable at a glance. Each square's size represents the return magnitude, creating an intuitive visual hierarchy. We originally included outlines on each square but removed them based on feedback for a cleaner, more streamlined appearance.

Insights:

The grid layout makes it easy to spot performance patterns across different time periods. Solar shows impressive 1-year returns (very large square) but much smaller returns over longer periods, suggesting recent gains rather than steady long-term growth. Oil & Gas Equipment & Services has consistently large squares across all time periods, indicating reliable performance over time. Software sectors show moderate, similar-sized squares throughout, reflecting stable returns. Traditional energy sources (Thermal Coal and Uranium) display smaller but consistent squares. This grid format quickly shows which industry and time period combinations perform best, and how different sectors behave over time.

Total Returns Trend per Industry (Over 10 Years)



Visualization 4 – Scatter Plot (Market Cap vs Total Return)

Type: Log-scale scatter plot with categorical encoding

Variables Mapped:

- x-axis (log scale): Market Capitalization
- y-axis: Total Return (5-Year, %)
- Point color: Sector (Energy vs Technology)
- Point shape: Group classification (Top 10 vs Bottom 10 performers)
- Text label: Company name

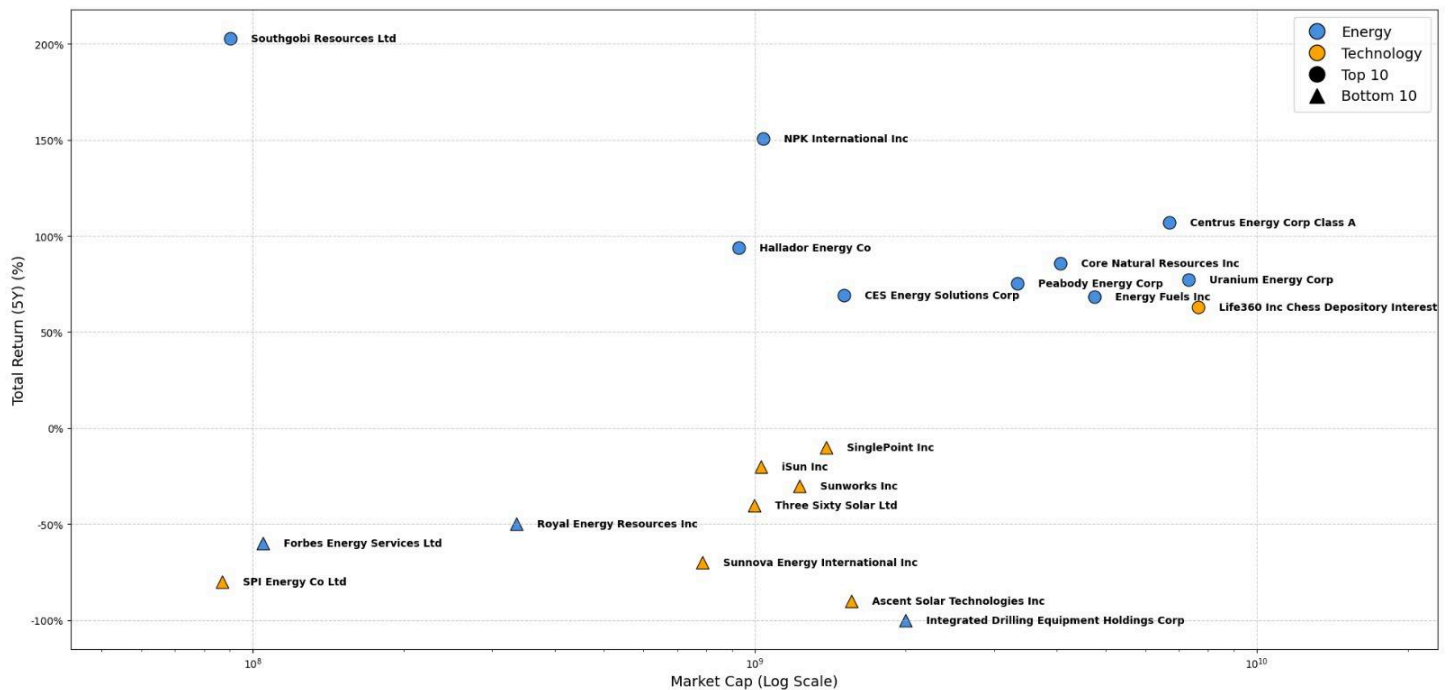
Refinement Process:

Added color and shape encodings for clearer categories, adjusted labels and axis limits for readability, and improved legend placement and size.

Insights:

Top performers are concentrated in the Energy sector with strong positive returns, while most underperformers are Technology firms with negative returns. Market cap shows no clear link to performance, and the color/shape encodings clearly separate sectors and performance groups.

Market Cap vs 5-Year Total Return (Energy & Technology - Top 10 vs Bottom 10)



Analysis and Discussion

Integrating the four visualizations reveals a cohesive narrative about the structure of global fund performance:

1. Return Dispersion by Style – Smaller-cap funds produce more extreme outcomes, emphasizing risk exposure and manager selection importance.
2. Regional Dominance – The U.S. overshadows other domiciles, shaping the global composition of investment opportunities.
3. Geographic Performance Ranking – Domicile correlates with average returns, influenced by market maturity and regional growth dynamics.
4. Sectoral Risk Profiles – Industry choice clearly separates performance patterns, with Energy companies showing strong positive returns while Technology firms exhibit greater volatility and more frequent negative outcomes.

Overall, the data demonstrates that market capitalization, geography, and industry act as intersecting axes of variability in long-term investment performance. By visualizing these dimensions simultaneously, we transformed abstract numerical data into an explanatory story that investors and analysts can interpret at a glance.

Individual Report

Danielle Siazon

As group liaison I was responsible for submitting milestone documents which meant I also had to consistently keep track of our progress. I actively communicated in the group chat about what needed to be done, I proposed having meetings in-person and online that accommodated everyone's availability and I made sure to be present in all of them.

My contributions included initiating milestone progress by creating documents, data exploration by running analysis in R and describing relationships to be studied while also taking up tasks that others were unable to do like cleaning the dataset. I produced graphs to be explored like the map in Milestone 2 and I also produced and refined the mosaic plot that is included in this final report and presented its purpose and insight during the presentation.

Lastly, the knowledge and materials I've learned in this class have been eye opening, especially the content about which graphs to use for certain variables, the right usage of colors, and knowing how to declutter visuals. I hope to incorporate these techniques into any future data projects which will definitely serve as an inspiration and guidance to how I can visually communicate data the right way to the right audience.

Ankita Tank

Throughout the project, I contributed actively to both coordination and analysis. I attended every meeting, led two sessions, and ensured the team met all deliverable deadlines. I proposed using Morningstar data, handling its extraction, and normalizing balancing the dataset.

I helped define the project's story examining investment performance, risk distribution, and style-based returns and drafted the initial outline that guided our analytical direction. I also conceptualized early visualizations such as scatter plots for performance, violin and boxplots for return distributions. Some of these ideas shaped our final explanatory figures.

This project taught me that visualization is both analysis and storytelling. I learned to iterate designs to uncover insights, apply normalization and scaling for comparability, and use effective color, typography, and layout for clarity. I also strengthened leadership and communication skills by coordinating progress and maintaining accountability.

Overall, this experience deepened my understanding of how thoughtful design and data integrity combine to communicate clear, impactful stories through visualization.

Taha

Throughout the project, I contributed to both our dataset decisions and the overall visuals (Colors, Typography, etc) of our work. I helped the team finalize our dataset and explored different types of visualizations during our early analysis. Even though some of the visuals I created were not included in the final report.

I also took the lead in designing our presentation template and made sure all graphs followed a consistent style, color treatment, and layout. This helped our final presentation feel clear, cohesive, and easy to follow. To support the group's workflow, I took initiative in scheduling some of our meetings and stayed active in communication to make sure everyone was aligned as we approached each milestone.

This project helped me understand the importance of story telling in data visualization. It also helped me grow more confident in collaborating with a team, giving and receiving feedback, and shaping work that feels both organized and impactful.

Sara Mohammed

For the milestones, I contributed by helping browse the dataset and refining our focus on financial data. My main contribution was creating the scatter plot for Milestone 3, which illustrated market capitalization and performance differences between Energy and Technology companies. I refined this visualization with feedback, suggested improvements to team visuals, and helped organize the presentation flow.

This project enhanced my understanding of data visualization, showing how scaling, color, and structure shape insight. It strengthened my confidence in turning data into clear, impactful stories.

Advaita Reddy Muduganti

I stayed active in group communication, attended meetings, and participated in decision-making by offering feedback. Presented the Introduction and Conclusion to ensure a strong opening and cohesive closing. This project strengthened my skills in exploratory visualization, narrative design, and collaborative workflow, giving me more confidence in shaping clear and meaningful stories from data.

Appendix - Code & Formative Analysis

1. Visualization 1 - Refer to attached file - violin_plot_visualization.R
2. Visualization 2 - Refer to attached file - mosaic_plot.R
3. Visualization 3 - Heat map Generated in Tableau
4. Visualization 4 - Refer to attached file - scatter_plot_visualization.py